Competitive Exam Practice 6: Process State Diagram, Schedulers, and Context Switch

GATE (Computer Science) Pattern

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Concepts That Click

Each question carries 1–2 marks. MCQ = single correct option; NAT = Numerical Answer Type. *All questions in this set are original, written in the GATE pattern; none is claimed to be a past-year question.*

Q1 *[Original, GATE-pattern]* **[MCQ, 1 mark]** In the five-state process model (`new`, `ready`, `running`, `waiting`, `terminated`), which of the following transitions is **NOT** possible?

(A) `running` → `ready`

(B) `waiting` → `ready`

(C) `waiting` → `running`

(D) `running` → `waiting`

Q2 *[Original, GATE-pattern]* **[MCQ, 1 mark]** The degree of multiprogramming is controlled primarily by the:

(A) short-term scheduler

(B) long-term scheduler

(C) dispatcher

(D) I/O-completion interrupt handler

Q3 *[Original, GATE-pattern]* **[MCQ, 1 mark]** Which scheduler removes a partially executed process from main memory to disk and later reintroduces it so that it continues where it left off?

(A) Long-term scheduler

(B) Short-term scheduler

(C) Medium-term scheduler

(D) Dispatcher

Q4 *[Original, GATE-pattern]* **[MCQ, 2 marks]** On a single-core system with other processes ready, which of the following events **necessarily** results in a context switch away from the current process?

(A) The process calls `getpid`

(B) A keyboard interrupt wakes another, waiting process

(C) The process calls `read` on a pipe that is currently empty

(D) The process executes an ordinary user-mode function call

Q5 *[Original, GATE-pattern]* **[MCQ, 2 marks]** Consider the following statements:

S1: Every context switch includes at least one switch between user and kernel mode.

S2: Every switch between user and kernel mode includes a context switch.

Which of the following is correct?

(A) S1 and S2 are both true

(B) S1 is true and S2 is false

(C) S1 is false and S2 is true

(D) S1 and S2 are both false

Q6 *[Original, GATE-pattern]* **[NAT, 1 mark]** A machine with 8 CPU cores has 50 processes. At some instant every core is executing a process, 11 processes are `ready`, 3 are `new`, and none is `terminated`. The number of processes in the `waiting` state is _____.

Q7 *[Original, GATE-pattern]* **[NAT, 1 mark]** A context switch takes 5 μs and each process runs for a time slice of 45 μs before being switched out. The percentage of CPU time spent on context switching is _____.

Q8 *[Original, GATE-pattern]* **[NAT, 2 marks]** A short-term scheduler takes 1 ms to select a process, after which the selected process runs for a time slice of q ms. The smallest value of q (in ms) for which the fraction of CPU time spent scheduling is at most 4% is _____.

Q9 *[Original, GATE-pattern]* **[NAT, 1 mark]** Consider the five-state process model and the following six transitions:

`new`→`ready`; `ready`→`waiting`; `running`→`ready`;
`waiting`→`running`; `waiting`→`ready`; `new`→`running`.

The number of these transitions that are legal is _____.

Q10 *[Original, GATE-pattern]* **[NAT, 2 marks]** Two CPU-bound processes `P1` and `P2` arrive at time 0 on a single core; `P1` is dispatched first. The CPU bursts are 6 ms for `P1` and 5 ms for `P2`. The time slice is 4 ms, a preempted process joins the tail of the ready queue, and switching time is negligible. Neither process does I/O. Counting only switches from one process to a *different* process, the number of context switches until both terminate is _____.

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